Jay Antonio Oliver

Applied Data Scientist | Data Engineering | Machine Learning

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Professional Summary

Data Scientist and Data Engineer with 6+ years of experience developing machine learning models, scalable ETL pipelines, and analytical solutions across environmental, energy, and scientific domains. Experienced with multi-million-record datasets, APIs, relational databases, and flat-file sources to support predictive modeling and decision-making.

Core Skills

Machine learning: scikit-learn, XGBoost, Random Forest, classification, regression, feature engineering, hyperparameter tuning, cross-validation, SHAP, pandas, NumPy.

Data engineering: Python, SQL, OOP, modular development, ETL, REST APIs, automation, relational databases, Git, Linux, Bash, Azure Cloud Storage.

Communication: stakeholder engagement, technical documentation, executive presentations, Power BI, matplotlib, Excel.

Professional Experience

Leidos — Geospatial Data Scientist

Morgantown, WV | Feb 2024 – May 2026

- Designed modular Python and SQL ETL pipelines integrating legacy files, databases, and operational datasets.
- Automated ingestion and transformation workflows, reducing processing effort by approximately 90%.
- Developed ML pipelines analyzing more than 7 million oil and gas well records to identify high-confidence orphaned-well candidates.
- Built Random Forest and XGBoost models for approximately 900,000 Appalachian Basin wells.
- Applied XGBoost regression to critical-mineral concentrations in coal ash datasets.

Baylor University — Lab Manager / Geospatial Data Scientist

Waco, TX | Apr 2023 – Feb 2024

- Led multidisciplinary research analytics and data science workflows.
- Developed Python ML and ETL pipelines integrating REST APIs, SQL databases, and flat files.
- Served as primary Python developer on land-use basemap work supporting a Scientific Data publication.

Geospatial Data Specialist

Oct 2021 – Apr 2023

- Built spatial data pipelines and ML workflows for transportation and environmental research.
- Integrated remote sensing, APIs, databases, and socioeconomic data for regional predictive modeling.

- Applied decision tree regression across Texas ecoregions and developed Python/OpenCV/Selenium camera-image ingestion workflows.

U.S. Geological Survey — Spatial Analyst

Remote | Jan 2021 – Jul 2021

- Developed Python and R workflows to process and visualize more than 20 years of environmental datasets, including temporal trends and land-cover change.

World Wildlife Fund — Data Scientist / Statistical Analyst

Remote / Washington, DC | Jan 2020 – Dec 2020

- Integrated geospatial, socioeconomic, and biodiversity datasets for long-term spatial trends spanning 2000–2017.
- Developed logistic regression workflows for spatial prediction using infrastructure, land use, and population variables.

NASA — Remote Sensing Data Scientist

Hampton, VA / Remote | Sep 2019 – Apr 2020

- Developed Python ETL workflows in Linux to integrate more than 900 multidimensional datasets across Peru and Central America.
- Processed large environmental time series for regional analysis and decision support.

Education

University of York — MSc, Environmental Economics and Environmental Management (Sep 2018)

University of South Florida — BS, Environmental Science and Policy (Dec 2014)